

# An Empirical Study of Price Dynamics in the Chinese Emission Trading Systems

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## Abstract

This paper provides an empirical analysis of price dynamics in China's emission trading systems (ETS), and compares the findings to global carbon market trends. Globally, carbon allowance prices tend to move with the economic cycle. In contrast, the Chinese carbon market shows different patterns. Both market-based risk factors and sector-specific influences—especially from the metals, aviation, and agriculture sectors—help explain carbon allowance returns. Regional trading systems in China show stronger negative relationships with energy prices, indicating greater sensitivity to rising input costs. In comparison, the national carbon market is more influenced by industry-level factors and reacts less to overall economic or financial conditions. These results highlight the distinctive features of China's ETS and the variation in price behavior across different regions.

**Keywords:** Carbon market, carbon allowance, emission trading system, climate finance

**JEL Classification:** G12, G15, Q54, Q58

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# 1 Introduction

With the rapid expansion of the global economy, carbon dioxide emissions have surged by 57.7%, increasing from 21.28 billion tons in 1990 to 33.57 billion tons in 2020.<sup>1</sup> In light of growing environmental, ecological, and economic concerns, the international community responded by establishing the United Nations Framework Convention on Climate Change (UNFCCC) in 1992—the first global treaty dedicated to tackling climate change. Building on this milestone, the Kyoto Protocol was adopted in 1997, introducing market-based mechanisms such as carbon trading, which allowed countries to exchange emission allowances and invest in emission-reduction initiatives in developing nations.

Carbon markets provide a direct means for economies to assign a price to carbon emissions, thereby internalizing the environmental costs of greenhouse gas (GHG) outputs. As a complementary policy instrument, carbon taxes not only generate substantial fiscal revenue but also enhance policy enforcement, boost public acceptance, and help mitigate concerns related to international competitiveness. Over the decade from 2013 to 2023, both the coverage and revenue of global carbon pricing instruments witnessed consistent growth. By 2022, a total of 73 carbon pricing mechanisms—including emissions trading systems and carbon taxes—were in operation, collectively covering 23% of global GHG emissions and facilitating transactions that surpassed \$100 billion.<sup>2</sup>

Achieving long-term climate goals requires maintaining or increasing carbon prices to incentivize low-carbon transitions. However, a range of economic, political, and environmental uncertainties continues to pose significant challenges. Between 2023 and 2024, several global developments—including unconventional monetary policies, rising geopolitical tensions, and the complexities of ongoing energy transitions—have disrupted carbon pricing mechanisms. These trends highlight the growing need for robust, science-based models of carbon pricing that can support

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<sup>1</sup>World Bank Carbon Pricing Dashboard: <https://carbonpricingdashboard.worldbank.org>

<sup>2</sup>World Bank (2022)

effective and sustainable climate policies.

Compared to traditional financial instruments, carbon-related products involve more complex pricing mechanisms. These mechanisms vary according to the type of instrument—such as carbon taxes, emissions trading systems, or voluntary offset markets—as well as in their application and institutional structures. This paper aims to provide a systematic overview of the key characteristics of carbon pricing tools, both globally and in China, in order to better understand their underlying market structures and dynamics.

Carbon prices are shaped by a diverse range of factors, including energy supply and demand, economic cycles, sector-level performance, and regional-specific conditions. This study investigates these drivers through empirical analysis of data from international carbon markets and China's nine carbon exchanges, offering a comprehensive foundation for future research in this area.

For China, the world's largest developing economy, establishing scientifically grounded carbon pricing is essential for aligning policy incentives with long-term sustainability goals. This empirical study contributes to that effort by providing quantitative evidence on the main drivers of carbon prices across various sectors. The findings carry implications for policymakers as well as academic researchers interested in environmental finance and climate policy.

Methodologically, the paper applies a series of empirical techniques, drawing on international carbon price data from the World Bank and Chinese ETS and financial market data from the CS-MAR database. Through time-series and regression analyses, the study examines the influence of macroeconomic, industrial, and regional variables on carbon prices.

This study makes several key contributions. First, it offers a systematic summary of the pricing and efficiency characteristics of different carbon instruments, helping to clarify the structure of the fragmented global carbon market. Second, it provides empirical evidence on the factors influencing both international and domestic carbon prices, incorporating environmental, industrial, and macroeconomic dimensions. Third, it disaggregates the Chinese ETS by regional exchange and

industry coverage, enhancing our understanding of the micro-level dynamics driving carbon price variation within China.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature on carbon markets and price formation. Section 3 outlines the data sources and empirical methodology. Section 4 presents the main regression results along with robustness checks. Section 5 concludes by summarizing the key findings and discussing their implications for future policy design.

## 2 Literature Review

In this section, I review the existing literature on previous studies of the carbon market.

### 2.1 Heterogeneity in Carbon Pricing Mechanisms

Carbon pricing varies in type, regulation, and effectiveness. According to the [World Bank \(2022, 2023\)](#), these mechanisms include emissions trading systems (ETS), carbon taxes, crediting schemes, results-based climate financing (RBCF), and internal pricing strategies. Carbon markets are typically categorized into compliance, voluntary, or offset markets.

ETS prices are often grounded in marginal abatement cost theory, but researchers such as [Montgomery \(1972\)](#) and [Luo \(2024\)](#) have pointed out that transaction costs and the allocation of surplus can lead to price deviations. Additionally, regional carbon pricing systems exhibit considerable variation due to overlapping jurisdictions and fragmented governance structures, as noted by [Jiang and Yu \(2024\)](#).

### 2.2 Implications of Mechanism Design

[Zhen et al. \(2023\)](#) and [Lin and Li \(2010\)](#) argue that hybrid ETS-tax schemes in developed countries exacerbate green trade barriers, limiting exports from developing nations. [Pan and Yang \(2024\)](#) highlights the regressive effects of carbon taxes, leading to welfare losses. [Wang and Yang \(2021\)](#)

finds that China's regional pilot ETS reduces linkages between the carbon and finance markets, while [Liu and Xu \(2023\)](#) reports positive spillover effects to neighboring provinces, based on difference-in-differences (DID) models.

### **2.3 Drivers of Carbon Prices in Chinese Market**

[Zhang et al. \(2021\)](#) argues that China's ETS, in contrast to cap-based systems used abroad, is based on output intensity and incorporates taxes and subsidies. [He \(2011\)](#) connects permit allocations to government emission targets. [Xu and Lien \(2024\)](#) documents significant spillovers between ETS returns and inflation. [Li \(2023\)](#) finds that exchange rates and industrial production impact carbon prices, while [Wang and Xu \(2021\)](#) explores the volatility driven by technological shocks and policy interventions.

### **2.4 Drivers of Carbon Prices in International Market**

[Seifert and Uhrig-Homburg \(2008\)](#) addresses seasonality in price models. [Burger et al. \(2008\)](#) identifies oil prices, political risk, temperature, development, and technological advancements as key drivers of carbon prices. [Meszade et al. \(2023\)](#) demonstrates that supply chain emissions influence firm-level pricing. [Liu and Xu \(2023\)](#) applies interval decomposition to quantify the impacts of the COVID-19 pandemic and the Ukraine war on the EU ETS.

## **3 Institutional Background and Data**

In this section, I begin by introducing the carbon market and its institutional background, followed by a description of the data used in the empirical analysis.

### **3.1 Carbon Markets and Products**

Carbon markets play a critical role in addressing climate change by putting a price on greenhouse gas emissions and incentivizing emission reductions. These markets have evolved into diverse sys-

tems, encompassing compliance-based trading schemes, voluntary offset mechanisms, and carbon taxation frameworks. Each approach varies in design, scope, and enforcement, reflecting the differing policy priorities and institutional capacities across jurisdictions. This section provides an overview of the major types of carbon pricing instruments: compliance markets such as Emissions Trading Systems (ETS), voluntary carbon markets (VCMs), and fiscal tools like carbon taxes and border adjustments. Understanding the structure and function of these instruments is essential to contextualize the empirical analysis that follows.

### **3.1.1 Compliance Carbon Markets**

Compliance carbon markets are government-regulated systems designed to limit greenhouse gas emissions by setting a cap on allowable emissions and enabling trading of emission allowances. These markets operate through two channels: the primary market, where emission permits are initially distributed—either freely allocated or auctioned—by regulatory authorities under a predefined cap; and the secondary market, where entities trade allowances among themselves based on their emission needs and compliance strategies.

As of 2023, there are 29 active compliance carbon pricing systems globally, covering nearly a quarter of global greenhouse gas emissions. Prominent examples include the European Union Emissions Trading System (EU ETS), UK ETS, California Cap-and-Trade Program, South Korea ETS, and emerging systems in Latin America and Southeast Asia ([World Bank, 2023](#)). These systems differ significantly in terms of coverage, stringency, allocation methods, and market maturity.

In China, the compliance carbon market architecture consists of both national and regional schemes. The national ETS, officially launched in July 2021, currently covers only the power generation sector, which accounts for over 40% of the country's emissions. It is now the world's largest ETS by volume of emissions covered. In parallel, eight provincial-level pilot ETSs (including Shenzhen, Beijing, Guangdong, and Shanghai) have been in operation since 2013, covering a broader set of

sectors such as manufacturing, transportation, and buildings. These pilots have served as valuable laboratories for policy experimentation, informing the design of the national system.

### **3.1.2 Voluntary Carbon Markets**

Voluntary carbon markets (VCMs) enable companies, organizations, and individuals to purchase carbon credits on a voluntary basis to offset their greenhouse gas emissions. These markets operate independently of regulatory requirements and are often driven by corporate sustainability goals, net-zero pledges, or consumer demand for environmentally responsible practices.

Carbon credits in VCMs are typically issued by certified projects that either reduce, avoid, or remove emissions from the atmosphere. These projects span a range of sectors, including renewable energy generation, energy efficiency, forestry, and carbon capture technologies. To ensure environmental integrity and credibility, projects are validated and verified by independent third-party standards such as Verra’s Verified Carbon Standard (VCS) and the Gold Standard.<sup>3</sup> These standards set requirements for additionality, permanence, monitoring, and verification.

Carbon credits in voluntary carbon markets can generally be classified into two categories. The first is emission reductions or avoidance, which refers to projects that prevent greenhouse gas emissions from being released into the atmosphere—for example, renewable energy installations, improved cookstoves, or energy efficiency improvements. The second is removals and sequestration, which includes projects that actively extract carbon from the atmosphere and store it, such as afforestation, reforestation, soil carbon enhancement, and emerging technologies like direct air capture.

### **3.1.3 Carbon Taxes**

Carbon taxes establish a predetermined price per unit of carbon dioxide emitted, providing a clear cost signal to emitters. They are often implemented alongside emissions trading systems (ETSs)

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<sup>3</sup>Verra was established in 2005 through a joint initiative by the Climate Group (CG), the International Emissions Trading Association (IETA), and the World Economic Forum (WEF). It oversees the VCS program, which is one of the most widely used standards in the voluntary market.

to enhance policy effectiveness and ensure broader emissions coverage. While China has not yet introduced a formal carbon tax, its 2022 Green Consumption Policy Framework outlines plans to explore carbon pricing instruments, including taxation, as part of its long-term climate strategy.<sup>4</sup>

In parallel, carbon border adjustment mechanisms (CBAMs) have gained traction as a way to prevent carbon leakage and ensure a level playing field in international trade. The European Union’s CBAM, scheduled for full implementation in 2026, will impose tariffs on imported carbon-intensive goods such as steel, cement, and aluminum, aligning trade policies with climate goals.

## **3.2 Carbon Markets Data**

### **3.2.1 Global Carbon Markets**

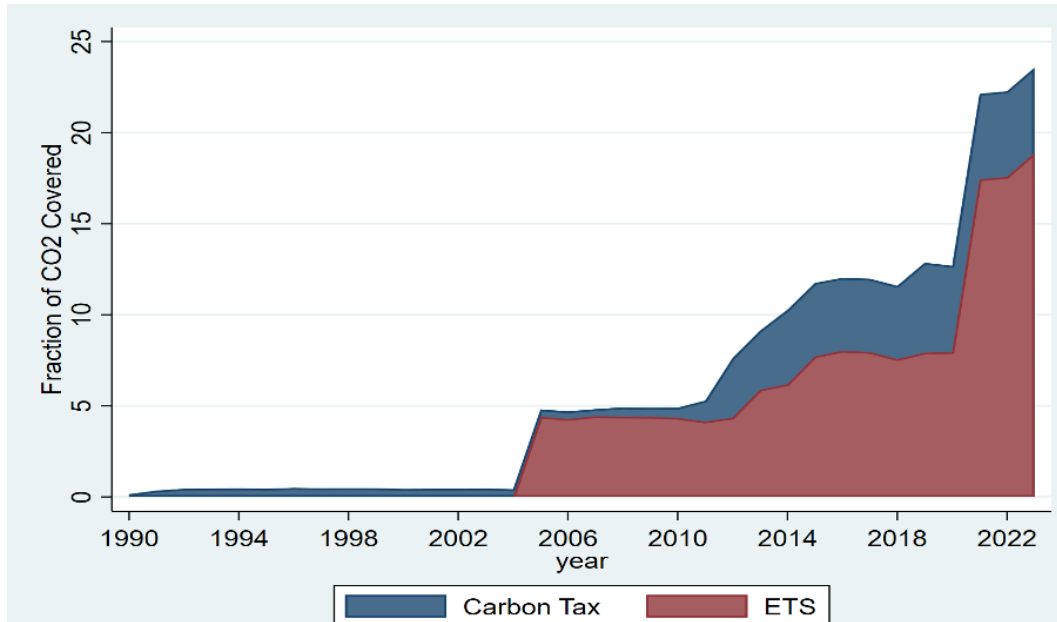
This study draws on data from the World Bank’s Carbon Pricing Dashboard<sup>5</sup> to analyze the evolution of carbon pricing mechanisms globally. The dataset provides annual information on the coverage and pricing of both carbon taxes and emissions trading systems (ETs), as well as country-level GDP data from 2005 to 2022.

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<sup>4</sup>State Council of China, 2022. “Implementation Plan for Promoting Green Consumption.”

<sup>5</sup>See World Bank source: [carbonpricingdashboard.worldbank.org](https://carbonpricingdashboard.worldbank.org)



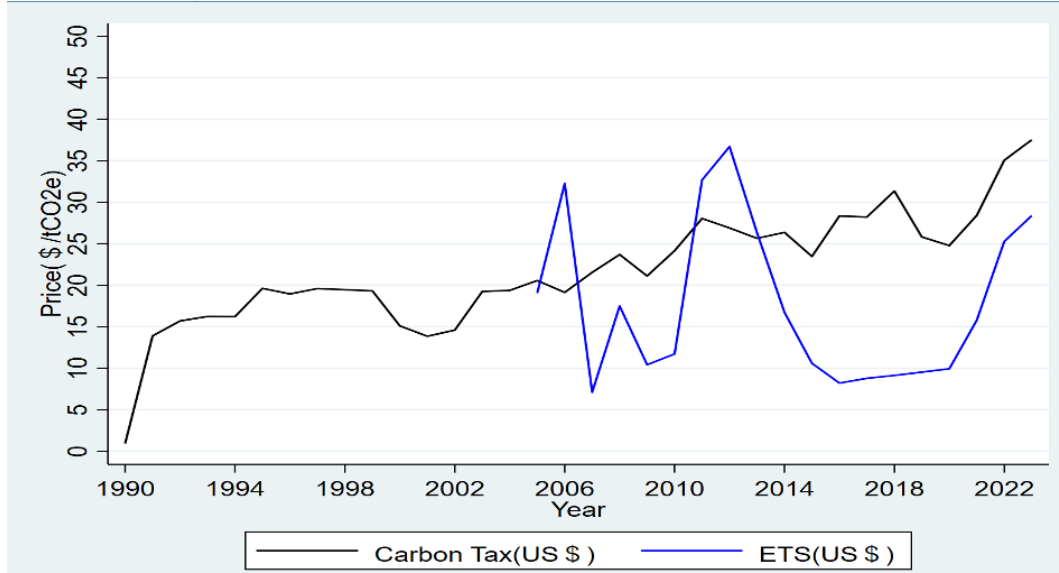


**Figure 1: Global Carbon Coverage by Product Type**

This figure illustrates the annual share of global CO<sub>2</sub> emissions covered by Emissions Trading Systems (ETS) in red and Carbon Taxes in blue from 1990 to 2022.

As is shown in Figure 1, ETSs have consistently covered a larger share of global greenhouse gas (GHG) emissions compared to carbon taxes. Global carbon pricing coverage increased significantly, from 4.41% of emissions in 2005 to 18.88% in 2022. Much of this growth occurred after 2019, driven by the launch of national trading platforms in China (2021) and Vietnam (2022).<sup>6</sup>

<sup>6</sup>China's national carbon market began operation in July 2021, while Vietnam's system was established in 2022. Trading activity data is available via the World Bank's Carbon Pricing Dashboard.



**Figure 2:** Global Carbon Price by Product Type

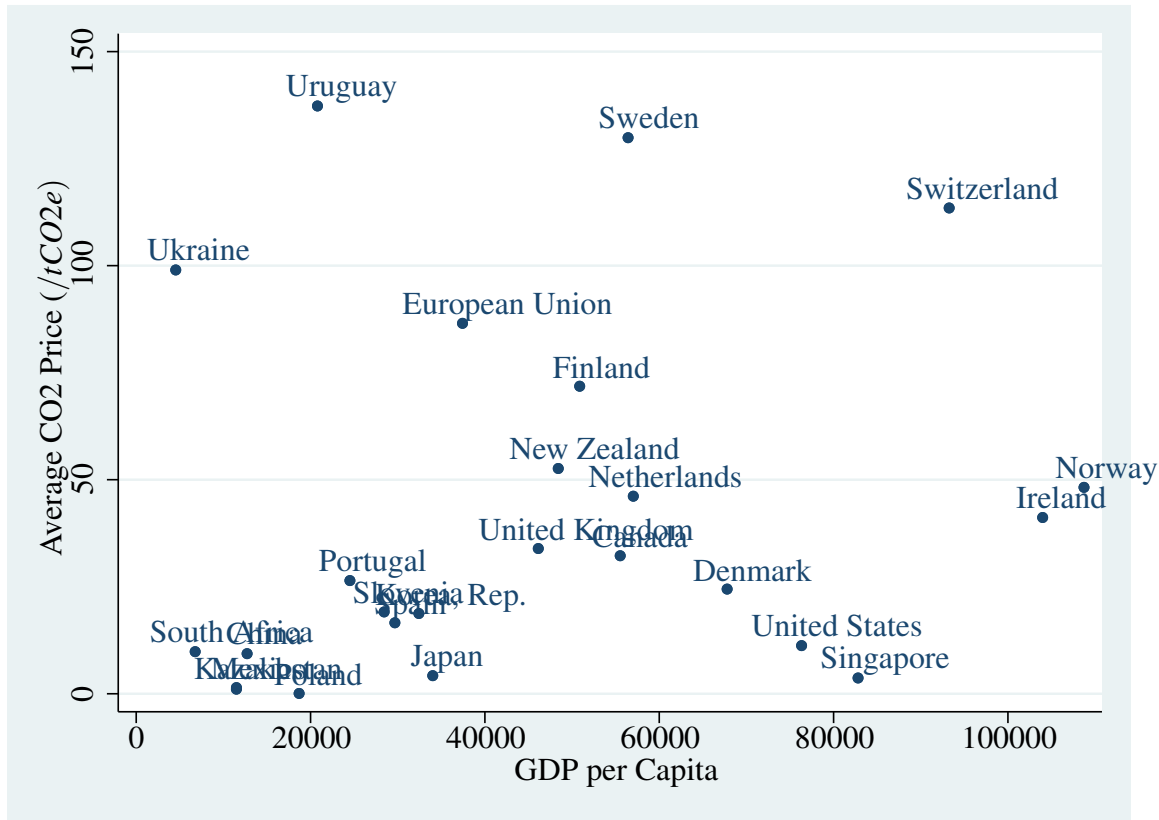
This figure illustrates the average carbon price under Emissions Trading Systems (ETS) in blue and Carbon Taxes in black from 1990 to 2022. Prices are calculated by weighting country-level carbon prices according to their respective GHG coverage.

As is shown in Figure 2, price levels are calculated by weighting country-level carbon prices by their GHG coverage.<sup>7</sup> ETS prices exhibit high volatility and fragmentation. Carbon tax prices are steadier.

<sup>7</sup>The weighted average carbon price is calculated as:

$$p_i = \sum_k w_k \cdot p_k$$

where  $w_k$  denotes the percentage share of global greenhouse gas emissions covered by mechanism  $k$  in year  $i$ , and  $p_k$  is the corresponding carbon price.



**Figure 3:** Carbon Prices Across Economies

This figure illustrates the relationship between the average carbon price and GDP per capita across different economies of year 2022.

Figure 3 presents a scatter plot of per-capita GDP versus carbon prices for 2022, revealing a positive correlation between the two variables. This suggests a pro-cyclical relationship, where carbon prices tend to increase with higher economic output. However, several notable outliers deviate from this trend. For instance, Ukraine exhibits relatively high carbon prices despite having a low GDP, which can be attributed to geopolitical disruptions and the associated economic and policy factors. Conversely, some developed economies, such as Norway, display lower carbon prices despite their high per-capita GDP. This anomaly may be explained by the significant penetration of clean energy technologies in these countries, which reduces the reliance on carbon-intensive energy sources and, consequently, the need for higher carbon prices. Such observations highlight the complexity of the relationship between economic performance and carbon pricing, which is influenced not only by economic factors but also by technological, political, and geopolitical considerations.

### 3.3 China’s Carbon Markets

Chinese Emissions Trading System (ETS) price indices and market data are sourced from the CSMAR database. This study utilizes daily trading data—including transaction prices and volumes—spanning the period from 2013 to 2024 across nine regional and national ETS platforms in China.

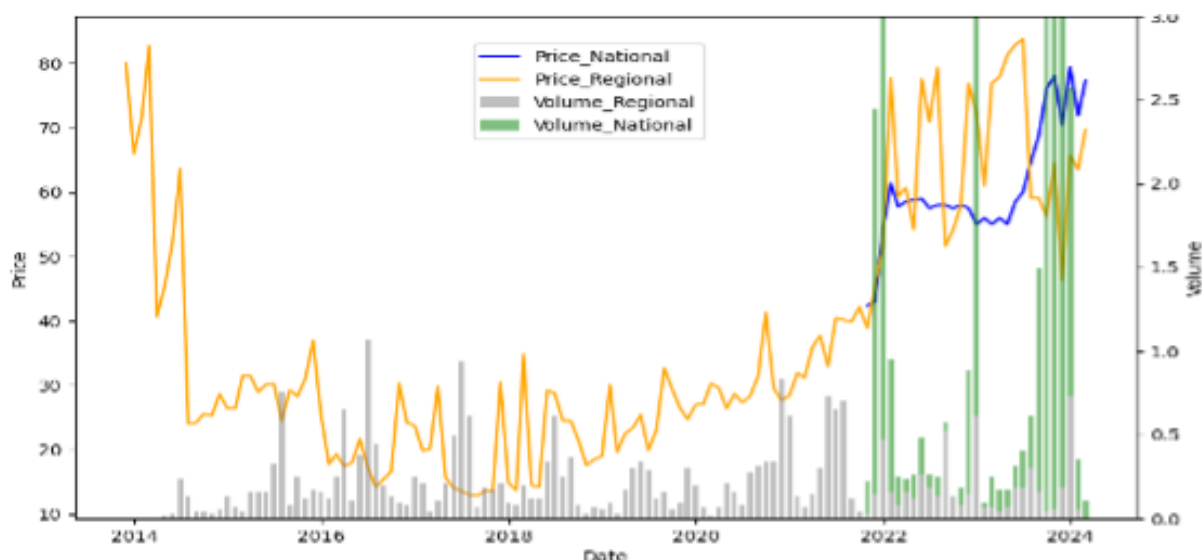
**Table 1:** Industry Coverage Across Regional ETS Platforms

This table reports the industry coverage of each ETS platform in China. There are nine ETS platforms, including eight regional platforms: Beijing, Tianjin, Shanghai, Chongqing, Hubei, Guangdong, Shenzhen and Fujian; as well as one National ETS platform.

| Industry           | Beijing | Tianjin | Shanghai | Chongqing | Hubei | Guangdong | Shenzhen | Fujian | National |
|--------------------|---------|---------|----------|-----------|-------|-----------|----------|--------|----------|
| Electric Power     | ✓       |         |          |           |       |           | ✓        |        | ✓        |
| Steel              |         | ✓       | ✓        | ✓         | ✓     | ✓         | ✓        | ✓      |          |
| Building Materials | ✓       | ✓       | ✓        | ✓         | ✓     | ✓         | ✓        |        |          |
| Petrochemical      | ✓       | ✓       | ✓        | ✓         | ✓     | ✓         |          | ✓      |          |
| Chemical Industry  |         | ✓       | ✓        | ✓         | ✓     |           | ✓        | ✓      |          |
| Non-ferrous Metals |         | ✓       | ✓        | ✓         | ✓     |           | ✓        | ✓      |          |
| Paper Making       |         |         |          | ✓         | ✓     | ✓         | ✓        | ✓      |          |
| Aviation           | ✓       | ✓       | ✓        |           |       | ✓         |          |        | ✓        |
| Transportation     | ✓       |         | ✓        |           |       |           | ✓        |        |          |
| Construction       |         |         | ✓        |           |       |           |          |        |          |
| Other Industries   | ✓       | ✓       | ✓        | ✓         | ✓     |           | ✓        | ✓      |          |
| Waste Treatment    |         |         |          | ✓         |       |           |          | ✓      |          |
| Food and Beverage  |         | ✓       |          | ✓         | ✓     |           |          |        |          |
| Service Industry   | ✓       |         | ✓        |           |       |           | ✓        |        |          |

As shown in Table 1, China’s regional ETS pilots—such as those in Shenzhen, Guangdong, and Hubei—exhibit broad sectoral coverage. These local platforms include energy-intensive industries

like electric power, steel, petrochemicals, aviation, and cement. In contrast, the national ETS, launched in 2021, currently covers only the electric power sector. Despite its limited sectoral scope, the national ETS has quickly become the largest in terms of trading volume, owing to its centralized structure and the significant emissions footprint of China’s power generation industry. The disparity in coverage across regional platforms reflects the experimental nature of China’s phased approach to carbon market development and the diverse industrial profiles of different provinces.



**Figure 4:** Monthly Price and Trading Volume of China’s ETS

This figure presents the monthly average carbon price and total trading volume across China’s emissions trading platforms. Data from the eight regional ETS pilots are aggregated to provide a comprehensive view. The average price is computed as a volume-weighted mean across the regional platforms, thereby capturing the relative influence of higher-volume markets on overall pricing dynamics.

As shown in Figure 4, carbon prices in China’s regional pilot exchanges—led by Shenzhen—were abnormally high before 2014, likely due to low market liquidity. Over time, as trading volumes increased across pilot platforms, carbon prices gradually rose. Notably, following the establishment of the national carbon emission trading scheme in 2021, the national exchange replaced local pilots as the dominant trading venue. However, it did not crowd out provincial exchanges; rather, it stimulated both price levels and transaction volumes in the pilots. This trend was particularly

pronounced between 2022 and 2023.

Trading volumes typically fell during mid-year months, while rising in late-year and early-year periods (November to February), accompanied by more pronounced price fluctuations. In late 2023, an unexpected spike in prices at the national exchange temporarily drove down the aggregate carbon price across platforms, a divergence that gradually faded in early 2024.

In summary, both transaction volume and pricing levels in China's carbon market show an upward trend. The national and regional markets display distinct characteristics. For empirical purposes, the 2018–2024 period provides a stable window in which carbon prices and volatilities appear smoother and more suitable for econometric analysis.

This study proposes several categories of explanatory variables that may influence carbon allowance prices, each supported by theoretical justification. The variables used in this study include: China Securities 500 Index (CSI500), S&P 500 Index (SPX), ChinaBond Composite Bond Index (Debt), CSI Metals Futures Index (Metal), CSI Agriculture Index (Farm), CSI Airline Industry Index (Airline), CSI Energy Futures Index (Energy), CSI Coal Futures Index (Coal), and Daily risk-free interest rate.

First, macroeconomic variables are considered, both domestic and international. Domestic indicators are divided into stock and bond indices: stock indices reflect the general business cycle, corporate earnings, and investment activity, while bond indices provide information on prevailing credit spreads and market liquidity. This distinction allows for a more nuanced understanding of the financial determinants of carbon asset pricing. International macroeconomic conditions may affect global commodity prices, foreign exchange rates, and other relevant financial variables.

Second, energy price variables are included. Carbon trading schemes primarily cover fossil fuel-based sectors, including coal, oil, natural gas, and LPG. However, the futures markets for these energy types in China vary in terms of inception dates and data availability. To ensure temporal consistency and avoid missing data, we employ a broad-based energy futures index and a coal

futures index—both of which exhibit continuous coverage and stable volatility over the sample period.

Third, we consider industry-specific performance as a potential driver. Based on Table 1, which outlines the industrial coverage across China’s carbon trading platforms, and Figure 4, which summarizes price and volume trends, we identify three representative sector indices: agriculture, metallurgy, and aviation. According to the International Energy Agency and Statista data, the industrial sector accounted for approximately 30% of China’s total emissions in 2020 (excluding energy production), while aviation and agriculture contributed 8% and 1%, respectively. Given their significance in carbon emissions and their relevance to ETS coverage, these sectors are included as explanatory variables.

Lastly, we incorporate regional market segmentation. Given structural differences between the national and local ETS platforms—including variation in industry coverage, trading volume, and regulatory design—we conduct separate regressions for the national and provincial ETS prices. This approach enables us to examine whether macroeconomic, energy, and sectoral factors exert different effects across distinct carbon market regimes.

## **4 Empirical Analysis**

### **4.1 Model Construction**

This section outlines the empirical framework developed to analyze the key drivers of excess returns on carbon allowances in China’s emissions trading system.

Following [Pedersen \(2023\)](#), the following linear regression model is constructed:

$$\begin{aligned}
Carbon_{i,t} = & \beta_1 \cdot CSI500_t + \beta_2 \cdot SPX_t + \beta_3 \cdot Debt_t \\
& + \beta_4 \cdot Metal_t + \beta_5 \cdot Airline_t + \beta_6 \cdot Farm_t \\
& + \beta_7 \cdot Energy_t + \beta_8 \cdot Coal_t + \varepsilon_t,
\end{aligned} \tag{1}$$

where  $Carbon_{i,t}$  denotes the excess return of carbon allowances on day  $t$  traded in platform  $i$ ,  $CSI500_t$  denotes the excess return of the CSI500 Index,  $SPX_t$  denotes the excess return of the S&P 500 Index,  $Debt_t$  denotes the excess return of the composite bond index,  $Metal_t$  denotes the excess return of the metals futures index,  $Airline_t$  denotes the excess return of the airline industry index,  $Farm_t$  denotes the excess return of the agriculture index,  $Energy_t$  denotes the excess return of the energy futures index,  $Coal_t$  denotes the excess return of the coal futures index

This model is designed to capture the impact of macroeconomic, sectoral, and energy market factors on the pricing of carbon allowances. The inclusion of both domestic and international indices allows for the examination of spillover effects and broader market sentiment.

Daily data are obtained from the CSMAR database for the period 2018–2023. Excess returns are computed by subtracting the corresponding risk-free rate from each index's daily return. All return series are synchronized to trading days across platforms and employed in the regression analysis.



**Table 2:** Summary Statistics of Variables

This table reports summary statistics for the variables used in the linear regression analysis. The data cover the period from 2018 to 2023 and are recorded at daily frequency. All variables are expressed as daily excess returns, computed by subtracting the risk-free rate from raw returns. Carbon refers to the excess return of carbon allowances; CSI500 and SPX represent the excess returns of the CSI 500 Index and the S&P 500 Index, respectively; Debt denotes the excess return of a composite bond index.

| Variable | Mean  | Std. Dev. | Min    | Max   | Nob   |
|----------|-------|-----------|--------|-------|-------|
| Carbon   | 0.15  | 3.80      | -17.03 | 19.78 | 3,954 |
| CSI500   | -0.05 | 1.09      | -5.49  | 8.98  | 3,954 |
| SPX      | 0.02  | 0.96      | -2.98  | 2.47  | 3,954 |
| Debt     | 0.02  | 0.04      | -0.08  | 0.09  | 3,954 |
| Metal    | 0.00  | 0.78      | -1.61  | 1.54  | 3,954 |
| Airline  | 0.04  | 1.41      | -2.83  | 2.77  | 3,954 |
| Farm     | -0.06 | 1.26      | -4.92  | 4.19  | 3,954 |
| Energy   | 0.02  | 1.30      | -2.43  | 2.55  | 3,954 |
| Coal     | 0.01  | 1.47      | -6.35  | 5.65  | 3,954 |

The descriptive statistics presented in Table 2 provide valuable insights into the behavior of the variables used in this study. The excess return on carbon allowances (*Carbon*) exhibits considerable variability, with a mean of 0.15 and a standard deviation of 3.80. This high level of dispersion suggests the presence of extreme values and reflects heterogeneity across different trading venues and industry sectors.

The CSI500 Index demonstrates relatively stable characteristics, with a mean close to zero (-0.05) and a modest standard deviation of 1.09. This indicates relatively low market volatility and a steady trend, potentially shaped by macroeconomic fundamentals.

Other variables, including the Coal Futures Index (*Coal*), the Agriculture Index (*Farm*), the S&P 500 Index (*SPX*), the Airline Index (*Airline*), and the Energy Futures Index (*Energy*), display

unique statistical features. For instance, the coal and agricultural indices have near-zero mean returns (0.01 and -0.06, respectively) but large standard deviations, implying significant intra-sector variation. The SPX also shows a mean close to zero and a relatively low standard deviation, suggesting a stable global market environment. In contrast, the airline and energy sectors show greater volatility, with standard deviations of 1.41 and 1.30, respectively. This may be driven by macroeconomic shocks and sector-specific factors such as fuel prices or demand cycles.

Finally, the *Debt* (ChinaBond Composite Index) and *Metal* (CSI Metals Futures Index) variables appear comparatively stable. The mean and standard deviation for *Debt* are 0.02 and 0.04, respectively, while those for *Metal* are 0.00 and 0.78. These values suggest relatively mature and liquid markets with lower levels of short-term price fluctuation.

## 4.2 Regression Results

This section presents the results of the regression analysis examining the determinants of excess returns on carbon allowances. To account for heterogeneity across trading venues, I estimate the empirical model separately for each emissions trading platform, including both regional and national markets. This approach allows for a comparative assessment of how various macroeconomic, financial, and sector-specific factors influence carbon pricing dynamics across different institutional and geographic contexts within China's ETS framework. To explore the relationship between carbon allowance returns and their potential determinants, I estimate a series of linear regressions. Each explanatory factor is first examined individually to assess its standalone impact. In the final specification, all variables are jointly included to evaluate their combined explanatory power.

**Table 3:** Full Sample Regression Results

This table presents the estimation results of Equation (1) using the full sample, which includes daily data from nine emissions trading platforms across China between 2018 and 2023.

|            | (1)     | (2)      | (3)    | (4)    | (5)    | (6)      | (7)    | (8)    | (9)      |
|------------|---------|----------|--------|--------|--------|----------|--------|--------|----------|
| CSI500     | 0.127** |          |        |        |        |          |        |        | 0.061    |
|            | (2.29)  |          |        |        |        |          |        |        | (0.75)   |
| Debt       |         | 3.817*** |        |        |        |          |        |        | 4.439*** |
|            |         | (2.82)   |        |        |        |          |        |        | (3.24)   |
| Metal      |         |          | 0.141* |        |        |          |        |        | 0.084    |
|            |         |          | (1.83) |        |        |          |        |        | (0.85)   |
| Energy     |         |          |        | 0.039  |        |          |        |        | -0.142   |
|            |         |          |        | (0.85) |        |          |        |        | (-1.57)  |
| Airline    |         |          |        |        | 0.073* |          |        |        | 0.056    |
|            |         |          |        |        | (1.71) |          |        |        | (1.08)   |
| Farm       |         |          |        |        |        | 0.136*** |        |        | 0.112*   |
|            |         |          |        |        |        | (2.83)   |        |        | (1.83)   |
| Coal       |         |          |        |        |        |          | 0.062  |        | 0.083    |
|            |         |          |        |        |        |          | (1.52) |        | (1.05)   |
| SPX        |         |          |        |        |        |          |        | 0.060  | 0.060    |
|            |         |          |        |        |        |          |        | (0.94) | (0.94)   |
| N          | 3,954   | 3,954    | 3,954  | 3,954  | 3,954  | 3,954    | 3,954  | 3,954  | 3,954    |
| Adj. $R^2$ | 0.01    | 0.02     | 0.01   | 0.01   | 0.01   | 0.02     | 0.01   | 0.01   | 0.04     |

Note: T-statistics are reported in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

This section presents linear regressions for each of the eight explanatory variables against the excess return of carbon allowances. The analysis is conducted using the full sample, which includes both national and regional (non-national) exchange data. The results are summarized in

Table 3.

Among the full sample regressions, five explanatory variables exhibit statistically significant coefficients when regressed individually: the CSI500 Index, the ChinaBond Composite Index, the Metals Futures Index, the Airline Industry Index, and the Agriculture Index. Notably, the bond and agriculture indices remain significantly and positively associated with carbon excess returns in the full model, at the 1% and 10% confidence levels, respectively. The significance of the remaining variables diminishes when all controls are included.

Compared to the relatively weak explanatory power of the domestic equity market in the full model (despite 5% significance in the individual regression), bond market conditions exert a more robust positive effect on carbon pricing. This may be due to the fact that an increase in bond yields signals higher credit costs for firms, thereby increasing the financing cost of carbon transactions and elevating returns on carbon allowances. In contrast, equity returns reflect broader market sentiment and investment activity, which may be less directly linked to transaction-level costs in the carbon market.

Among sectoral variables, agriculture is the only industry that retains significance in the full regression (10% level). One plausible explanation is that agricultural activities encompass a broader range of emission sources compared to other industries such as aviation or metallurgy. These include biological respiration, fertilizer and pesticide use, and carbon emissions from food processing. While the airline industry shows marginal significance in the individual regression (10% level), this disappears in the full model. This suggests that the original signal may be driven by macroeconomic activity rather than firm-level operating performance.

**Table 4:** Sub-Sample Regression Results (Regional ETSs)

This table presents the regression results of Equation (1) based on daily data from the eight regional emissions trading systems (ETSs) in China.

|            | (1)               | (2)                | (3)             | (4)             | (5)             | (6)                | (7)              | (8)              | (9)                |
|------------|-------------------|--------------------|-----------------|-----------------|-----------------|--------------------|------------------|------------------|--------------------|
| CSI500     | 0.136**<br>(2.17) |                    |                 |                 |                 |                    |                  |                  | 0.062<br>(0.68)    |
| Debt       |                   | 4.660***<br>(3.02) |                 |                 |                 |                    |                  |                  | 5.230***<br>(3.39) |
| Metal      |                   |                    | 0.123<br>(1.39) |                 |                 |                    |                  |                  | 0.044<br>(0.38)    |
| Energy     |                   |                    |                 | 0.047<br>(0.87) |                 |                    |                  |                  | -0.172*<br>(-1.67) |
| Airline    |                   |                    |                 |                 | 0.077<br>(1.54) |                    |                  |                  | 0.071<br>(1.10)    |
| Farm       |                   |                    |                 |                 |                 | 0.148***<br>(2.72) |                  |                  | 0.123*<br>(1.77)   |
| Coal       |                   |                    |                 |                 |                 |                    | 0.079*<br>(1.69) |                  | 0.121<br>(1.35)    |
| SPX        |                   |                    |                 |                 |                 |                    |                  | 0.0715<br>(0.98) | 0.072<br>(0.98)    |
| N          | 3,402             | 3,402              | 3,402           | 3,402           | 3,402           | 3,402              | 3,402            | 3,402            | 3,402              |
| Adj. $R^2$ | 0.01              | 0.02               | 0.01            | 0.01            | 0.01            | 0.02               | 0.01             | 0.01             | 0.05               |

Note: T-statistics are reported in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

The regression results for the regional (non-national) carbon exchange in Table 4 reveal that four explanatory variables exhibit some degree of statistical significance in the individual regressions: the CSI500 Index, the ChinaBond Composite Index, the Agriculture Index, and the Coal Futures

Index. Among these, the bond and agriculture indices show the highest significance, with confidence levels exceeding 99%.

In the full model, the ChinaBond Index and the Agriculture Index remain significant, at the 1% and 10% levels, respectively—consistent with the full-sample results. Additionally, the Energy Futures Index shows a significant negative effect at the 10% level. This may reflect the mechanism through which rising energy prices increase firms' production costs, thereby reducing demand for emissions-intensive activities and suppressing carbon allowance returns.

Overall, the drivers of regional carbon allowance pricing align with those observed in the full sample, including sensitivity to bond market-implied transaction costs and the agriculture sector's emissions profile. However, the negative impact of energy prices is more pronounced in the regional sample, likely due to the structural differences in sectoral coverage and trading frequency between national and regional exchanges (see Table 1 and Figure 4). Accordingly, the following section focuses on the national exchange to further investigate its distinct pricing determinants.

**Table 5:** Sub-Sample Regression Results (National ETS)

This table presents the regression results of Equation (1) based on daily data from the national emissions trading system (ETS) in China.

|            | (1)    | (2)     | (3)     | (4)    | (5)    | (6)    | (7)     | (8)     | (9)     |
|------------|--------|---------|---------|--------|--------|--------|---------|---------|---------|
| CSI500     | 0.060  |         |         |        |        |        |         |         | 0.042   |
|            | (0.73) |         |         |        |        |        |         |         | (0.37)  |
| Debt       |        | -1.467  |         |        |        |        |         |         | -1.053  |
|            |        | (-0.78) |         |        |        |        |         |         | (-0.55) |
| Metal      |        |         | 0.229** |        |        |        |         |         | 0.269** |
|            |        |         | (2.35)  |        |        |        |         |         | (2.17)  |
| Energy     |        |         |         | 0.000  |        |        |         |         | 0.111   |
|            |        |         |         | (0.00) |        |        |         |         | (0.85)  |
| Airline    |        |         |         |        | 0.056  |        |         |         | -0.015  |
|            |        |         |         |        | (1.07) |        |         |         | (-0.22) |
| Farm       |        |         |         |        |        | 0.057  |         |         | 0.041   |
|            |        |         |         |        |        | (0.84) |         |         | (0.49)  |
| Coal       |        |         |         |        |        |        | -0.040  |         | -0.207* |
|            |        |         |         |        |        |        | (-0.71) |         | (-1.71) |
| SPX        |        |         |         |        |        |        |         | -0.003  | 0.008   |
|            |        |         |         |        |        |        |         | (-0.04) | (0.10)  |
| N          | 552    | 552     | 552     | 552    | 552    | 552    | 552     | 552     | 552     |
| Adj. $R^2$ | 0.01   | 0.01    | 0.08    | 0.02   | 0.01   | 0.01   | 0.01    | 0.02    | 0.04    |

Note: T-statistics are reported in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

The regression results in Table 5 reveal that the determinants of carbon allowance pricing in the national exchange differ substantially from those in the full sample and the regional exchange sample. In the full model, macroeconomic indicators such as the ChinaBond Composite Index,

CSI500 Index, and S&P 500 Index show no statistically significant influence.

The most significant factor appears to be the Metals Futures Index, which shows a strong positive effect on carbon returns at the 1% confidence level. In addition, the Coal Futures Index exhibits a statistically significant negative effect at the 10% level in the full model. This may reflect a mechanism wherein rising metal prices enhance the profitability of metallurgical firms, thereby increasing their demand for emissions allowances and pushing up allowance prices. Conversely, higher coal prices raise production costs for carbon-intensive firms, which may reduce their emissions output and subsequent demand for carbon permits, thereby lowering prices.

Unlike the full and regional samples, the national exchange sample shows almost no sensitivity to broad macroeconomic conditions. Instead, it appears more responsive to industry-specific factors and energy market conditions. This divergence could be attributed to differences in sectoral coverage (see Table 1) and regional characteristics of the exchange itself, as the national carbon trading platform is headquartered in Shanghai, China.

### 4.3 Robustness Check

To address potential nonlinearity and scale effects that may arise from using arithmetic returns, we perform a robustness check by modifying the original model (1). Specifically, we replace the percentage excess return of carbon allowances with log excess return, defined as  $r_t = \ln(P_t/P_{t-1})$ .

The adjusted regression model is specified as:

$$\begin{aligned} Carbon\_LN_t = & \beta_1 \cdot CSI500_t + \beta_2 \cdot SPX_t + \beta_3 \cdot Debt_t \\ & + \beta_4 \cdot Metal_t + \beta_5 \cdot Airline_t + \beta_6 \cdot Farm_t \\ & + \beta_7 \cdot Energy_t + \beta_8 \cdot Coal_t + \varepsilon_t \end{aligned} \quad (2)$$

where  $Carbon\_LN_t$  represents the daily log excess return of China's carbon allowances.



**Table 6:** Robustness Check

This table presents the regression results of Equation (2), based on daily carbon allowance data from nine emissions trading systems (ETSs) in China, covering the period from 2018 to 2023.

|            | (1)     | (2)     | (3)    | (4)    | (5)    | (6)      | (7)    | (8)    | (9)      |
|------------|---------|---------|--------|--------|--------|----------|--------|--------|----------|
| CSI500     | 0.130** |         |        |        |        |          |        |        | 0.062    |
|            | (2.34)  |         |        |        |        |          |        |        | (0.77)   |
| Debt       |         | 3.180** |        |        |        |          |        |        | 3.788*** |
|            |         | (2.35)  |        |        |        |          |        |        | (2.76)   |
| Metal      |         |         | 0.140* |        |        |          |        |        | 0.083    |
|            |         |         | (1.82) |        |        |          |        |        | (0.84)   |
| Energy     |         |         |        | 0.039  |        |          |        |        | -0.144   |
|            |         |         |        | (0.84) |        |          |        |        | (-1.59)  |
| Airline    |         |         |        |        | 0.071* |          |        |        | 0.057    |
|            |         |         |        |        | (1.66) |          |        |        | (1.02)   |
| Farm       |         |         |        |        |        | 0.138*** |        |        | 0.113*   |
|            |         |         |        |        |        | (2.87)   |        |        | (1.85)   |
| Coal       |         |         |        |        |        |          | 0.062  |        | 0.083    |
|            |         |         |        |        |        |          | (1.52) |        | (1.05)   |
| SPX        |         |         |        |        |        |          |        | 0.061  | 0.061    |
|            |         |         |        |        |        |          |        | (0.97) | (0.97)   |
| N          | 3,954   | 3,954   | 3,954  | 3,954  | 3,954  | 3,954    | 3,954  | 3,954  | 3,954    |
| Adj. $R^2$ | 0.01    | 0.01    | 0.01   | 0.01   | 0.01   | 0.02     | 0.01   | 0.01   | 0.03     |

Note: T-statistics are reported in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

The robustness regression is performed on the same 3,954 daily observations as the baseline model. As shown in Table 6, switching from percentage to log returns leads to a slight reduction in the model's explanatory power, with marginally lower  $t$ -statistics across variables compared to Ta-

ble 3.

Nevertheless, the key findings remain consistent: the ChinaBond Composite Index and the Agriculture Index retain their statistical significance at the 1% and 10% levels, respectively. This consistency reinforces the reliability of the original specification and suggests that the relationship between these macro/sectoral factors and carbon allowance prices is robust to return transformation.

## 5 Conclusion

This study empirically investigates the determinants of carbon allowances prices in China's emissions trading system (ETS), providing insight into how macroeconomic, financial, and sector-specific variables influence carbon pricing dynamics. By constructing a multivariate regression framework and employing daily data from 2018 to 2023 across nine trading platforms, the analysis provides robust evidence on the systemic and sectoral factors shaping the carbon market.

The findings reveal several key conclusions. First, among macroeconomic indicators, the ChinaBond Composite Index emerges as the most consistent and statistically significant determinant of carbon returns across full-sample, regional, and robustness specifications. This suggests that bond market conditions—likely reflecting broader monetary policy and credit availability—play a crucial role in shaping carbon allowance prices, possibly through their influence on transaction and compliance costs.

Second, the Agriculture Index exhibits a stable and positive association with carbon returns across most model specifications. This result indicates the carbon-intensive nature of the agricultural sector and its sensitivity to emission cost structures. By contrast, other sectoral indices such as Energy, Coal, and Airline show limited or context-dependent significance, implying that sector-specific linkages to carbon pricing are heterogeneous and vary across different institutional settings.

Third, the national carbon market exhibits distinct pricing behavior compared to regional markets. While macroeconomic factors are more influential in regional exchanges, the national market is primarily driven by industry-specific indicators, notably the Metals Futures Index and the Coal Futures Index. This distinction points to structural differences in sectoral coverage, trading activity, and policy implementation across platforms.

Finally, the robustness check using log returns confirms the stability of the core findings. The consistency of key variable significance and directionality indicates that the results are not artifacts of return transformation or model specification.

The results highlight the complex nature of carbon pricing in China's ETS, driven by both macroeconomic forces and sector-specific dynamics. While financial indicators such as bond yields play a consistent role, industry-level factors—particularly in agriculture and heavy industry—also significantly influence allowance returns. This suggests that effective analysis and policy design must consider both national economic conditions and the structural characteristics of emitting sectors.

The study also offers practical policy implications. Introducing carbon taxes and regulated offset markets can improve price coverage, liquidity, and support for low-carbon initiatives like afforestation. Greater inter-provincial coordination—through unified rules and standardized data—can enhance market efficiency and transparency. Finally, aligning China's ETS with international frameworks would reduce market fragmentation and help avoid emerging carbon trade barriers.

Future studies could explore socio-economic variables (e.g., education, income inequality) as additional determinants of cross-country carbon price variation. Analyzing these could refine understanding of behavioral and institutional factors shaping carbon pricing.

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